NewsLens: Midterm Technical Milestone

Actual progress snapshot as of 12 October 2026 | Yi Tan (2312260)

Progress summary

The data audit, reproducible split, three lexical classifiers, two self-trained Word2Vec comparisons, local web demonstration, TF-IDF temperature scaling, review threshold, additive term explanation, deletion diagnostic, and automated system tests have been implemented. This document is an early snapshot, not a claim that the course-designated midterm date has arrived.

The Flask service processes input locally and provides class probabilities, review advice, and terms supporting the selected class over the runner-up. Three lexical models are selectable online; the other methods are offline comparisons. The interface reads stored experiment metrics rather than displaying handwritten scores.

Method	Accuracy	Macro-F1	Runtime (s)
binary	0.9878	0.9725	7.7
frequency	0.9905	0.9774	20.4
tfidf	0.9913	0.9803	6.9
glove	0.9887	0.9756	38.0
w2v_ag	0.9835	0.9637	88.0
w2v_nyt	0.9896	0.9769	138.5
bert	0.9748	0.9439	7313.6

Preliminary system evidence

The fitted TF-IDF temperature is 0.3387; ten-bin test ECE changes from 0.0167 to 0.0048. The validation-selected threshold is 0.9526. The system records per-example predictions, metrics, confusion matrices, hyperparameter trials, and split identifiers so each summary can be checked.

Ten automated tests cover shared split membership, tokenization, invalid and oversized input, out-of-vocabulary handling, probability normalization, exact explanation reconstruction, and successful page and API responses. They verify the implemented workflow; they are not a production security assessment.

Challenges and interpretation

The sports class dominates the corpus, so macro-F1 and per-class errors must accompany accuracy. Exact duplicate text occurs across the original train and test split. The main comparison retains the fixed course split, while a supplemental TF-IDF analysis excludes eight repeated test texts.

The neural model has a 64-WordPiece limit, while lexical classifiers can use full articles. A length audit finds that all 1,152 test articles exceed the neural limit, so performance differences cannot be attributed solely to architecture. A local CPU and large public model downloads also slow complete neural experiments.

Keyword contributions are algebraic effects on a linear class-score difference. They can be verified exactly, but are not causal explanations or evidence that an article is true. Calibration and accepted-accuracy estimates remain specific to this dataset and split.

Plan adjustment and next steps

The initial model comparison has been expanded to include negative log likelihood, expected calibration error, accepted coverage, an exact contribution check, a deletion diagnostic, and duplicate sensitivity. The validation rows were split so fitting the temperature and selecting the threshold use disjoint subsets.

Before the formal milestone and final presentation, update this snapshot with all completed required model results, retest the packaged demo from a clean environment, inspect every PDF page, and prepare a direct eight-minute explanation. Multi-seed, event-grouped, and out-of-domain evaluations are future extensions unless recorded artifacts demonstrate they were run.

Individual contribution and tool use

Sole member: Yi Tan (2312260). Responsibilities cover the full workflow: problem framing, data, modeling, calibration, explanation, integration, tests, reporting, and presentation. Libraries and pretrained weights provide the standard algorithms. Codex assisted with drafting, execution, debugging, and artifacts. The specific boundary between prior work and project work is documented in the separate contribution statement.